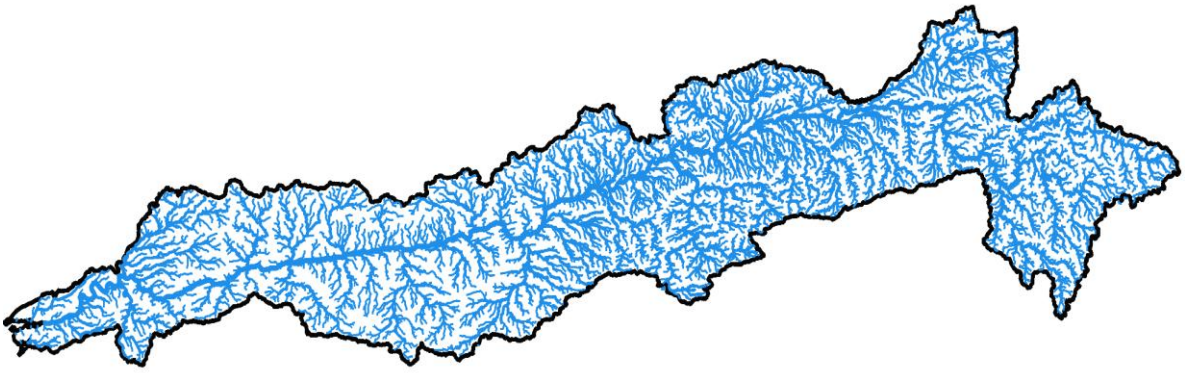
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Management Plan for Domestic Wastewater



June 2026

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The Centre for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the centre serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

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www.cganga.org

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PREFACE

This report presents the comprehensive Management Plan for Domestic Wastewater in the Narmada River Basin. The Narmada River Basin supports millions of people across central and western India and is an important ecological and economic region.

Due to increased urbanization and population in the basin, managing domestic effluents and emerging pollutants has become increasingly important for protecting the ecosystem.
This report assesses the status of domestic wastewater generation, treatment infrastructure, model identification for the Narmada Basin, solutions for urban and rural areas, and management.

This report identifies the key challenges and gaps and provides recommendations for sustainable domestic wastewater management practices. The report is intended to serve as a reference for policymakers, regulatory authorities, stakeholders, and researchers working towards the conservation and sustainable management of the Narmada River Basin.

Centre for Narmada River Basin
Management and Studies (cNarmada)
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Abbreviations and Acronyms

Abbreviation	Full Form
ABR	Anaerobic Baffled Reactor
AMRUT	Atal Mission for Rejuvenation and Urban Transformation
ASP	Activated Sludge Process
BOD	Biochemical Oxygen Demand
DEWATS	Decentralized Wastewater Treatment Systems
FSSM	Faecal Sludge and Septage Management
FSTP	Faecal Sludge Treatment Plant
HHs	Households
MBBR	Moving Bed Biofilm Reactor
MBR	Membrane Bioreactor
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act
MLD	Million Litres per Day
MoHUA	Ministry of Housing and Urban Affairs
MPPCB	Madhya Pradesh Pollution Control Board
NMCG	National Mission for Clean Ganga
PPP	Public-Private Partnership
SBM-U	Swachh Bharat Mission – Urban
STP	Sewage Treatment Plant
ULB	Urban Local Body
WTP	Water Treatment Plant
ZLD	Zero Liquid Discharge

1 Introduction & Objectives

The Narmada River is one of the most significant west-flowing river systems in India, traversing the states of Chhattisgarh, Madhya Pradesh, Maharashtra, and Gujarat before draining into the Arabian Sea. The basin spans a large area and supports a population of approximately 20,936,840, making it a critical socio-economic and ecological region. The population distribution within the basin is spatially heterogeneous, with 8,603,425 people in the upper basin, 9,126,886 in the middle basin, and 3,206,529 in the lower basin (Demography Report; cNarmada, 2025).

Domestic Wastewater is one of the largest sources of water pollution in the Narmada River Basin. Many residents live in areas that have an inadequate or no sewer system. As a result, many residents have untreated wastewater discharged into their neighborhood drainage systems, tributaries, etc., and ultimately into the river. Unsurprisingly, many rural/tribal districts have little or no access to central treatment facilities. For example, some districts report that 100 percent of their sewage is untreated. Not surprisingly, most of these districts lack sewage treatment plants or sewers. Even though many districts have some form of treatment infrastructure, efficiency is often lacking.

Because of differences in population and infrastructure across the upper, middle, and lower basins, a cumulative pollutant load exists throughout the river system. Untreated wastewater from upper basin districts progressively degrades water quality as you move further down the river. Therefore, developing a basin-wide analysis using both demographics and infrastructure will be required to create an effective management strategy. In addition to developing an analysis, a basin-wide comprehensive domestic wastewater management plan will be needed for the Narmada River Basin. Such a plan will need to consider how population density varies by district, how much wastewater is generated per person/district, and gaps in treatment infrastructure, so targeted methods can be implemented to promote sustainable management of water resources.

The objective of this report is to analyze the existing infrastructure of the Narmada basin in terms of domestic wastewater management, identification of gaps in the treatment facility, and sewage generation.

The existing infrastructure and sewerage networks for water and wastewater in the Narmada Basin consist of Water Treatment Plants (WTPs), Sewage Treatment Plants (STPs), and Pumping systems. All these infrastructures support water supply and wastewater management. collectively. There is spatial variation of infrastructure across the upper, middle, and lower basin regions, which ultimately reflects the population, urbanization, and institutional capacity.

1.1 Water Treatment and Supply Infrastructure

There is significant regional variability in the water treatment and distribution infrastructure within the Narmada Basin. Most of the ULBs throughout the Upper, Middle, and Lower Basins rely on the Narmada River as their principal water supply, as shown in *Table 1*. Infrastructure development is best illustrated in the Upper Basin, where a couple of large cities, such as Jabalpur, have an advanced treatment capacity of 298 MLD and a distribution network of 1870 km. On the other hand, there are smaller towns in this area with limited treatment capacity and medium-sized distribution networks. Several towns located here include Amarkantak, Mandla, Saikheda and even some ULBs that still have active construction going on.

A more evenly dispersed and balanced infrastructure was seen in the Middle Basin area. These areas included large urban areas such as Narmadapuram (24 MLD & 225 km Pipeline) and Barwani (12 MLD & 150 km Pipeline). The district of Khargone had two separate town-sized treatment facilities connected via a short pipeline system. Dhar also had a combination of separate treatment plants serving each town, with different pipeline lengths. However, it should be noted that towns like Bherunda, Nemawar, Barwah, and Sendhwa are still under construction. In contrast, the Lower Basin provided many examples of fully developed infrastructure. One example of this is Bharuch with its higher treatment capacity (70.4 MLD) and extensive distribution networks (510km). Larger systems, such as the Broach Region (Vadodara), showed a much greater capacity (180 MLD) but lack detailed information. Some of the smaller towns in this area, like Ankleshwar and Chhota Udaipur, had limited data available.

Table 1: Basin-wise Status of Water Treatment and Distribution Infrastructure in the Narmada Basin, including Source, Treatment Capacity (MLD), Pipeline Network Length,

and Project Status across Urban Local Bodies. *Source: Hamari Jeewan Rekha Narmada by https://mpurban.gov.in/* ***(2024)***

	Name of the ULB	District	Water Distribution Project			
			Source of water	**Water treatment plant capacity (MLD)**	**Water distribution pipeline length (KM)**	**Status of the project**
Upper Narmada Basin	Amarkantak	Anuppur	Narmada	2.09	59	Under Construction
	Dindori	Dindori	Narmada	4.9	35.2	Completed
	Mandla	Mandla	Narmada	2.09	59	Under Construction
	Jabalpur	Jabalpur	Narmada	298	1870	Completed
	Bhedaghat	Jabalpur	Narmada	1.2	24	Completed
	Narshingpur	Narshingpur	Narmada	13.1	120	Completed
	Saikheda	Narshingpur	Narmada	3	33.127	Under Construction
Middle Narmada Basin	Narmadapuram	Narmadapuram	Narmada	24	225	Completed
	Budhni	Sehore	Narmada	4.5	40	Completed
	Bherunda (Nasrullaganj)	Sehore	Narmada	13.68	196	Under Construction
	Nemawar	Dewas	Narmada	2.21	45.33	Under Construction
	Omkareshwar	Khargone	Narmada	5.62	23	Completed
	Barwah	Khargone	Narmada	4	50	Under Construction
	Sanawad	Khargone	Narmada	7.5	55	Completed
	Mandleshwar	Khargone	Narmada	3.74	22	Completed
	Maheshwar	Khargone	Narmada	6	32	Completed
	Dhamnod	Dhar	Narmada	5.1	79	Completed

	Dharampuri	Dhar	Narmada	3.4	46	Completed
	Anjad	Khargone	Narmada	4	28.5	Completed
	Barwani	Barwani	Narmada	12	150	Completed
	Sendhwa	Barwani	Narmada	9	91.25	Under Construction
Lower Narmada Basin	Ankleshwar	Bharuch	Narmada	Not Available	Not Available	Ongoing Operations
	Bharuch	Bharuch	Narmada	70.4	510	Completed
	Chhota Udaipur	Vadodara	Narmada	Not Available	Not Available	Planning Stage
	Broach (Vadodara)	Vadodara	Narmada	180	Not Available	Ongoing Expansion
	Rajpipla	Narmada	Narmada	6.0	42	Completed

1.2 Domestic Sewage Treatment Infrastructure

The existing water and sewage treatment systems vary widely from one district to another within the Narmada Basin, in terms of accessibility and available capacity. In the upper part of the Narmada Basin, most districts, i.e., Balaghat, Betul, Damoh, Dindori, Raisen, and Umaria, do not have Sewage Treatment Plants (STPs) in operation. This indicates a significant infrastructure gap. Even though some districts already have treatment capacity, as evidenced by cities such as Jabalpur (146 MLD) and Sagar (43 MLD), the large number of towns in these districts that still lack STPs suggests incomplete coverage.

Among the districts in the upper basin, Jabalpur appears to have the most functional STP infrastructure. Thus, compared with the rest of the districts in the upper basin, the sewerage system in Jabalpur appears to be among the best developed.

In contrast to the upper basin, many districts in the middle part of the Narmada Basin (i.e., Bhopal 177 MLD, Dewas 48 MLD, and Burhanpur 25.5 MLD) currently have significant available treatment capacity. However, some districts in this area (i.e., Dhar, Khandwa, Khargone, and Barwani) have much less than is needed relative to demand, and many projects are now being built or planned in this region. The lower part of the Narmada Basin (Alirajpur) currently lacks treatment capacity, but proposals are underway to build new facilities to support future development, as shown in *Table 2*.

Overall, the number of towns needing STPs far exceeds the number that have working STPs across all parts of the Narmada Basin. Consequently, many areas continue to discharge raw wastewater into surface waters. While numerous proposed and ongoing projects suggest that infrastructure expansion is occurring at some level in response to emerging needs, the existing conditions clearly point to an urgent need to accelerate implementation/development and to provide greater levels of service (i.e., full coverage) to meet increasing demands for wastewater management in the Narmada Basin.

Table 2: District-wise Status of Sewage Treatment Infrastructure in the Narmada Basin, including Existing Capacity, Under-Construction and Proposed STP Capacity, and Coverage of Towns Requiring Facilities.

*Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

	District	**No. of Towns STPs installed in the district**	**No. of Towns needing STPs**	**Total available Treatment Capacity (MLD) in the district**	**Under Construction STP Infrastructure Capacity (MLD)**	**Proposed STP Infrastructure Capacity (MLD)**
Upper	Anuppur	Nil	10	1.3		11.19
	Balaghat	0	4	0		6.5
	Betul	0	2	0		7.95
	Chhindwara	-	-	0	28	10.4
	Damoh	0	1	0		7.6
	Dindori	0	2	0	3.8	1
	Narmadapuram	0	4	0	21	10.7
	Jabalpur	Jabalpur-6, Bhedaghat-5	7	146	9.1	11.8
	Katni	Nil	3	0	24.5	8.7

	District	No. of Towns STPs installed in the district	No. of Towns needing STPs	Total available Treatment Capacity (MLD) in the district	Under Construction STP Infrastructure Capacity (MLD)	Proposed STP Infrastructure Capacity (MLD)
	Mandla	Nil	1	0.6		2
	Narsinghpur	1	5	9		13.05
	Raisen	1	12	0	2.1	20.8
	Sagar	1	17	43	17.4	4.45
	Seoni	-	-	0		23.8
	Umaria	0	5	0		7.3
Middle	Alirajpur	0	1	0		0.75
	Barwani	0	2	0	17.5	
	Betul (Middle portion)	0	2	0		7.95
	Bhopal	18	1	177		
	Burhanpur	1	None	25.5		3.7
	Dewas	3	1	48		9
	Dhar	1	7	2		5.8
	Harda	None	Need STP in all ULB	0	NA	NA
	Indore	NA	NA	0		
	Jhabua	-	1	0	5	2.6
	Khandwa	2	2	1		1.75
	Khargone	2	7	21.35		4.8
	Raisen (Middle portion)	1	12	0	2.1	20.8
	Sehore	1	9	12	4.2	12.45

Lower	Alirajpur (Lower)	0	1	0		0.75

1.3 Sewer Network Infrastructure

Urban Local Bodies (ULBs) in the Narmada Basin exhibit considerable variability at the regional and urban scales in terms of household connections to sewerage systems, as shown in *Table 3*. While the Upper Basin includes several large-scale cities with extensive sewer connections (i.e., Jabalpur -- 103,942 connected HHs; Narsinghpur-- 15,000 HHs; Mandla-- 13,200 HHs), other smaller urban areas in this region are characterized by limited sewerage availability (Amarkantak; Bhedaghat; Saikheda). Conversely, the Middle Basin demonstrates a more dispersed distribution of sewerage connections. Urban areas that demonstrate a relative increase in household connections to sewerage systems (Narmadapuram: 30,350 HHs; Sendhwa: 12,000 HHs) can be compared with those that exhibit lower connection rates (Barwah, Sanawad, Maheshwar, Anjad, Nemawar, Dharmapuri). The overall picture suggests that a greater proportion of small- and mid-sized towns in both the upper and middle basins currently use some form of non-sewered sanitation system. This indicates a strong rationale for expanding existing sewer systems and/or implementing FSSM systems to provide adequate wastewater treatment for all residents in these municipalities.

Table 3: Basin-wise Households Connected to Sewerage Systems in Urban Local Bodies of the Narmada Basin. *Source: Hamari Jeewan Rekha Narmada (https://mpurban.gov.in/)*

Narmada Basin	Name of the ULB	District	Households Connected to Sewerage System
Upper	Amarkantak	Anuppur	1,518
	Dindori	Dindori	4,500
	Mandla	Mandla	13,200
	Jabalpur	Jabalpur	103,942
	Bhedaghat	Jabalpur	1,084
	Narshingpur	Narshingpur	15,000
	Saikheda	Narshingpur	2,478
Middle	Narmadapuram	Narmadapuram	30,350
	Budhni	Sehore	2,804

	Bherunda (Nasrullaganj)	Sehore	4,274
	Nemawar	Dewas	1,800
	Omkareshwar	Khargone	2,400
	Barwah	Khargone	5,082
	Sanawad	Khargone	8,257
	Mandleshwar	Khargone	2,460
	Maheshwar	Khargone	4,577
	Dhamnod	Dhar	6,500
	Dharmapuri	Dhar	2,300
	Anjad	Khargone	5,488
	Barwani	Barwani	10,380
	Sendhwa	Barwani	12,000

The Narmada Basin's sewerage network coverage gap demonstrates the need for improved infrastructure development in smaller cities and rural municipalities. Only a few of the upper basin's districts, including Anuppur, Dindori, Narmadapuram, Jabalpur, Narsinghpur, and Sagar, have partially underground sewerage networks. However, the majority of the district's towns remain without sewerage. Several districts within the middle basin include Khargone (four ULBs), Dhar (two ULBs), and Dewas, Barwani, and Sehore (each with one ULB), all of which have partial underground sewerage systems. Alirajpur is representative of the lower Basin, where most towns lack sewerage networks. Details are shown in *Table 4*.

Table 4: District-wise Distribution of Partial Underground Sewerage Networks and Towns without Sewerage Infrastructure across the Upper, Middle, and Lower Narmada Basin.
*Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

Narmada Basin	District	No. of ULBs have a partial underground sewerage network	No. of Towns not having a sewerage network
Upper	Anuppur	1	9
	Balaghat	Nil	Nil

	Betul	0	7
	Chhindwara	-	-
	Damoh	0	6
	Dindori	1	1
	Narmadapuram	1	7
	Jabalpur	2	7
	Katni	Nil	Nil
	Mandla	Nil	1
	Narsinghpur	1	7
	Raisen	0	-
	Sagar	1	17
	Seoni	-	-
	Umaria	0	5
Middle	Alirajpur	0	3
	Barwani	1	8
	Betul	0	7
	Bhopal	BMC -56%	1
	Burhanpur	1	2
	Dewas	1	12
	Dhar	2	9
	Harda	NIL	4
	Indore	NA	NA
	Jhabua	0	0
	Khandwa	1	None
	Khargone	4	5
	Raisen	0	-
	Sehore	1	8
Lower	Alirajpur	No	3

2 Demand vs Capacity

2.1 Identified Infrastructure Gaps

The demand vs. capacity assessment of sewage treatment infrastructure across the Narmada basin clearly indicates a widespread and significant mismatch between domestic wastewater generation and available treatment capacity, highlighting a critical challenge for sustainable water management in the region *Table 5*. This imbalance represents one of the most serious challenges facing the long-term sustainability of water resource use throughout the Narmada River Basin. At present, nearly all districts in the upper-middle and lower basins of the Narmada River Basin have substantially inadequate infrastructure to treat their respective volumes of domestic wastewater. As shown in the table of deficits, the upper basin has some of the highest deficit values (e.g., Jabalpur (-103 MLD), Narmadapuram (-56.87 MLD), Betul (-49.28 MLD), and Katni (-34.56 MLD)) due to the combined effects of increasing volumes of municipal wastewater being discharged into sewers and/or treatment systems, along with the inadequacy of treatment system capacity. Similarly, some of the largest urban and/or semi-urban districts within the middle basin, including Bhopal (-69.94 MLD), Dhar (-45.9 MLD), Khandwa (-34.91 MLD), and Barwani (-19 MLD), among others, exhibit large treatment deficits. Finally, the lower basin's representative district, Alirajpur, has no treatment facility for domestic discharge.

When planned (under-construction/proposed) STP capacity is included, then although treatment deficits decrease from their levels when only existing/operational STP capacities were considered; however, several districts still appear to be significantly deficient in meeting future wastewater treatment needs, including those in Jabalpur (-82.1 MLD), Dhar (-40.1 MLD), Betul (-41.33 MLD), and Khandwa (-33.16 MLD).

Overall, the analysis suggests that the proposed and under-construction STPs will reduce deficits in treatment capacity; however, there is a need for proper investment plans for infrastructure and enhancements in treatment capacity, as well as the implementation of a basin-wide planning strategy to ensure effective and sustainable management of wastewater resources.

Table 5: District-wise Assessment of Sewage Generation, Existing Treatment Capacity, Infrastructure Gap, and Projected Capacity Expansion in the Upper, Middle, and Lower Narmada Basin, including Expected Gap after Completion of Ongoing and Proposed STP Projects. *Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

Narmada Basin	District	Sewage Generation in the district (MLD)	Existing Sewage Treatment Capacity (MLD)	Present Infrastructure Gap (MLD)	Future Capacity (Existing + Under Construction+ Proposed) Infrastructure (MLD)	Gap After Completion of the Projects (MLD)
Upper	Anuppur	13.27	1.3	-11.97	12.49	-0.78
	Balaghat	19	0	-19	6.5	-12.5
	Betul	49.28	0	-49.28	7.95	-41.33
	Chhindwara	54.43	0	-54.43	38.4	-16.03
	Damoh	13.89	0	-13.89	7.6	-6.29
	Dindori	2.64	0	-2.64	4.8	2.16
	Narmadapuram	56.87	0	-56.87	31.7	-25.17
	Jabalpur	249	146	-103	166.9	-82.1
	Katni	34.56	0	-34.56	33.2	-1.36
	Mandla	Nil	0.6	0.6	2.6	2.6
	Narsinghpur	16	9	-7	22.05	6.05
	Raisen	28	0	-28	22.9	-5.1
	Sagar	71.07	43	-28.07	64.85	-6.22
	Seoni	20.8	0	-20.8	23.8	3
	Umaria	8.53	0	-8.53	0.75	-7.78
Middle	Alirajpur	4	0	-4	0.75	-3.25
	Barwani	19	0	-19	17.5	-1.5

	Betul (portion)	49.28	0	-49.28	7.95	-41.33
	Bhopal	246.94	177	-69.94	177	-69.94
	Burhanpur	33	25.5	-7.5	29.2	-3.8
	Dewas	53.31	48	-5.31	57	3.69
	Dhar	47.9	2	-45.9	7.8	-40.1
	Harda	14.67	0	-14.67	1.7	-12.97
	Indore	62.16	NA	NA	NA	NA
	Jhabua	7.36	0	-7.36	7.6	0.24
	Khandwa	35.91	1	-34.91	2.75	-33.16
	Khargone	34.97	21.35	-13.62	26.15	-8.82
	Raisen	28	0	-28	22.9	-5.1
	Sehore	27	12	-15	28.65	1.65
Lower	Alirajpur	4	0	-4	0.75	-3.25

Table 6 : Domestic Wastewater Generation in the Lower Narmada Basin (Gujarat)

Source: Bharuch Nagar-palika & Ankleshwar Nagar-palika

District	Domestic Waste Generation
Chota Udepur ULB	4MLD
Rajpipla ULB	4.5MLD
Ankleshwar ULB	7.5MLD
Bharuch ULB	28MLD

Table 6 presents the estimated domestic wastewater generated by major Urban Local Bodies (ULBs) in the Lower Narmada Basin. These data provide an overview of domestic wastewater generation and serve as an important indicator for planning wastewater management and pollution control strategies.

Table 7: Sewage Treatment Plant (STP) Capacity in the Lower Narmada Basin (Gujarat)Source*: Bharuch Nagar-palika & Ankleshwar Nagar-palika*

District	Sewage Treatment Plant

Chota Udepur ULB	5MLD
Rajpipla ULB	5.5MLD
Ankleshwar ULB	14MLD
Bharuch ULB	29.3MLD

Table 7 summarizes the installed sewage treatment plant (STP) capacities available in major urban centres of the Lower Narmada Basin. Identified Models for the Narmada Basin.

The wastewater generation, infrastructure, and treatment gaps analysis indicates that a uniform wastewater management approach is not acceptable due to significant spatial heterogeneity. It is necessary to combine elements from various approaches, including centralized, decentralized, hybrid, and nature-based, with each type of dependent on factors such as population density, level of urbanization, amount of sewerage generated, and availability of infrastructure. Further field-level investigation also supports the application of these approaches, especially in areas with high levels of urbanization (such as Jabalpur), where they may be most effective at intercepting and treating wastewater, as demonstrated by their use in wastewater management.

2.2 Centralized Interception and Treatment Model for Urban Areas (Jabalpur Model)

For districts that are very densely populated and highly urbanized, e.g., Jabalpur, Dhar, and Khargone, a centralized intercept-and-treat model will be the best choice. In this model, nallahs (drainage) are diverted and intercepted for treatment at both central and decentral STPs (Sewer Treatment Plants) before being discharged into rivers. The results of our field observation in Jabalpur demonstrate the efficiency of this type of model. According to the report entitled "Integrated Urban Water and Wastewater Management Report" for Jabalpur municipality, all major nallahs were intercepted/tapped, and sewage was sent to treatment plants before discharge. Riverbank survey and inspection from a boat confirmed that there are no large quantities of untreated sewer discharges entering the Narmada River within the limits of the urban area. *Figure 1 and Figure 2* show Centralized STP a.) Lalpur 34 MLD STP & b.) and Decentralized STPs Jabalpur a.) Gwarighat STP 0.55 MLD & b.) Kathonda 32 MLD STP.

Therefore, it can be concluded that an effective drainage intercept and sewage collection system has been implemented.

The Jabalpur model is characterized by:

- Interception and diversion of major drains before river discharge
- Combination of large centralized STPs (e.g., 34 MLD plants at Lalpur and Kathonda) and decentralized small STPs near ghats (0.5-1 MLD scale)
- Use of multiple treatment technologies such as SBR, MBBR, ASP, and MBR, ensuring operational flexibility
- Integration of fecal sludge treatment (FSTP) for septage management
- Continuous monitoring and verification of outfalls



Figure 1: *Picture of Centralized STP a.) Lalpur 34 MLD STP & b.) Kathonda 32 MLD STP, Source: Clicked by cNarmada, IIT Indore team during visit. (30/01/2026)*



Figure 2: *Picture of Decentralized STPs Jabalpur a.) Gwarighat STP 0.55 MLD & b.) Kharighat STP 1 MLD, Source: Clicked by cNarmada, IIT Indore team during visit. (30/01/2026)*

2.3 Decentralized Wastewater Treatment Model for Rural and Low-Density Areas

The use of DEWATS provides an economical and viable option for rural and sparsely populated districts (Dindori, Mandla, Alirajpur, and Seoni), where conventional sewerage system infrastructure does not exist or cannot be implemented due to constraints. Such areas have low-density populations, many scattered small communities with limited economic and technical capabilities that make centralizing sewerage treatment systems unfeasible (Capodaglio, 2017). Decentralized treatment eliminates the need for lengthy sewer pipes from the generation point to the treatment point; at each location, wastewater is treated through simple, low-cost technologies, including settling tanks, anaerobic baffled reactors (ABRs), filters, etc., followed by polishing units, such as constructed wetlands or ponds. A primary goal of these technologies is the removal of organic material and particulate material via natural biochemical reactions.

In developing countries where infrastructure is poor or absent, nature-based treatment systems are appropriate because they require very little power and maintenance while providing stable water quality. In addition, since the treated effluent is reusable on-site for irrigation and other purposes, its reuse is beneficial for agricultural communities. In terms of feasibility, DEWATs are inexpensive, made with readily available community materials, relatively easy to manage at the community level (Massoud et al., 2009), and can be easily scaled up as needs grow.

2.4 Hybrid Wastewater Management Model for Medium Towns

A hybrid model for medium-scale districts with partially available sewerage infrastructure. In this approach, both centralized STP in core urban areas and decentralized systems in peripheral zones can be used. The characteristics of Hybrid Wastewater Management Models are:

- Centralized treatment for dense urban cores
- Decentralized units for peri-urban and unsewered areas
- Phased infrastructure development
- Improved cost-efficiency and scalability

This model ensures that areas without sewer connectivity are also covered under wastewater treatment systems.

2.5 Fecal Sludge and Septage Management (FSSM) Model

Many households in the Narmada Basin have relied on on-site sanitation systems, like septic tanks, pit toilets, and soak pits. These systems generate large amounts of fecal sludge and septage when there is no sewer network. Therefore, FSSM becomes a crucial part of an integrated wastewater management strategy for the basin. Many households in the Narmada Basin use on-site sanitation systems, including septic tanks, pit latrines, and soak pits. On-site sanitation systems generate large volumes of fecal sludge and septage due to the lack of a complete sewage system.

2.5.1 Components of the FSSM Model

The FSSM model is designed to ensure safe collection, transport, treatment, and reuse/disposal of fecal sludge, as shown in *Figure 3*. It comprises the following key components:

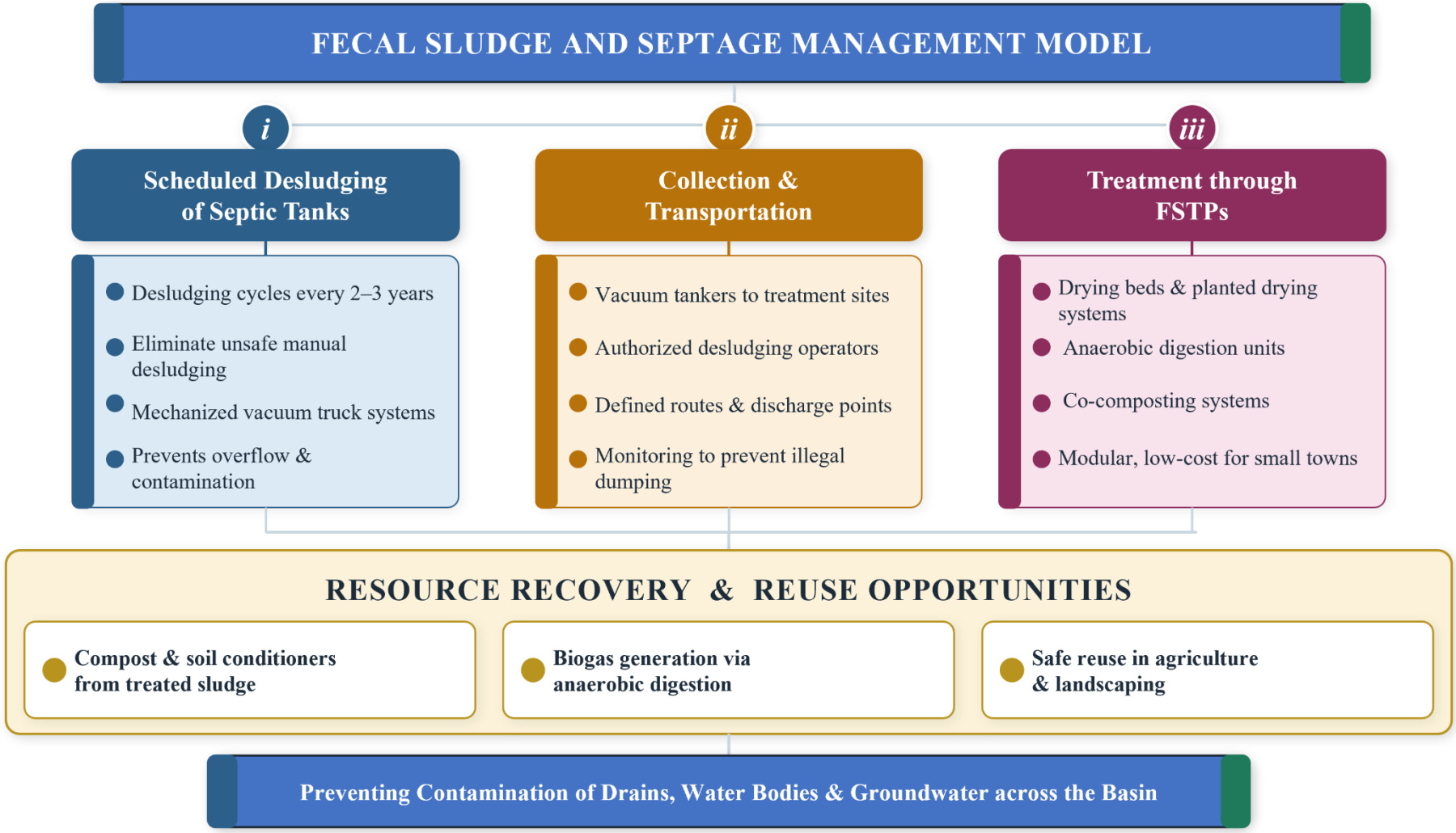


Figure 3: Fecal Sludge and Septage Management (FSSM) Model for the Narmada Basin, Components, Process Flow, and Resource Recovery (Source: - NITI Aayog & NFSSM Alliance, 2021)

2.5.2 Case Implementation: Jabalpur and Narmadapuram

The presence of operational Fecal Sludge Treatment Plants (FSTPs) in cities such as Jabalpur and Narmadapuram clearly shows that the FSSM model is both practical and scalable within the Narmada Basin, as shown in *Figure 4*. Field observations have shown that integrating septage management into existing wastewater infrastructure enables effective treatment of both sanitary and non-sanitary sewage. At Jabalpur, the FSTP is well linked to the urban sanitation system, therefore, allowing proper treatment of sludge from onsite sanitation systems so that it does not end up draining into the drain system or the Narmada River. It supports the existing sewerage infrastructure and helps strengthen wastewater management in Jabalpur. The FSTP in Narmadapuram has an approximate capacity of 0.9 MLD and was developed specifically to treat fecal sludge generated by onsite sanitation systems. Site observations also showed a defined treatment process and an established composting practice that converts treated sludge into useful products supporting resource recovery:

- FSSM systems are essential for bridging gaps in sewerage infrastructure
- Dedicated FSTPs can effectively handle high strength septage loads
- Resource recovery options such as composting can improve economic viability



Figure 4: *Fecal sludge treatment plant a.) Jabalpur & b.) Narmadapuram*

Source: Clicked by cNarmada, IIT Indore team during visit.(27/01/2026)

3 Resource Recovery Plan

The rising volumes of untreated or partially treated wastewater throughout the Narmada Basin present a substantial opportunity to transition from traditional waste-based treatment processes to an integrated resource recovery approach. Contemporary wastewater management approaches embrace the notion of "wastewater as a resource," in that treated effluent, biosolids (sludge), and other byproducts can be utilized to produce water, energy, and nutrient products. This approach aligns with circular economy principles and has been widely advocated in global studies on sustainable sanitation and water reuse (van Loosdrecht & Brdjanovic, 2014; Verstraete et al., 2009). In the context of the Narmada basin, where a substantial infrastructure gap exists, integrating resource recovery strategies can simultaneously reduce pollution loads and improve the economic viability of wastewater treatment systems.

3.1 Wastewater Reuse for Agriculture and Non-Potable Applications

A substantial area in the Narmada Basin is comprised primarily of agriculture and peri-urban settlement, providing an opportunity to use reclaimed water from treated wastewater through various means. Water scarcity and reliance on agriculture in districts like Dhar, Khargone, and Betul; therefore, wastewater can serve as a supplemental irrigation resource.

Secondary- and tertiary-treatment wastewater can also be utilized in cities such as Jabalpur and Bhopal, for non-drinking applications, i.e., landscape irrigation, industrial cooling, construction/maintenance of urban parklands, and other forms of greenbelts. Separating potable and non-potable use within dual water distribution systems will significantly reduce the use of fresh water.

3.2 Energy Recovery through Biogas and Waste-to-Energy Systems

Sewage sludge generated from treatment plants represents a valuable resource for energy recovery through anaerobic digestion. High loads in areas like Jabalpur, Bhopal, and Khargone offer significant potential for biogas production from large volumes of sewage. Biogas produced by anaerobic digestion of sludge contains large amounts of methane and therefore has many applications, including power generation and heat generation, or can be upgraded to compressed biogas (CBG) for use as vehicle fuel.

The study also states that it is feasible to make WWTPs energy-neutral or even energy-positive through optimized sludge digestion and the implementation of energy recovery systems (Verstraete et al., 2009). The application of an integrated approach using national schemes such as the SATAT scheme may improve the financial viability of these systems.

3.3 Nutrient Recovery and Biosolids Management

Domestic wastewater is a rich source of nitrogen, phosphorus, and organic carbon, all of which are essential for agricultural productivity. The Recovery of Nutrients from Domestic Wastewater Has Been Extensively Studied as a Sustainable Alternative to Conventional Fertilizer Production (Loosdrecht & Brdjanovic, 2014). Biosolids Can Be Processed into Compost or Bio-Fertilizers and Applied to Agricultural Fields After Meeting Safety Standards.

The Use of Treated Biosolids in Rural Districts Such as Dindori, Mandla, and Seoni, Which Are Dominated by Agriculture, Could Have a Significant Positive Impact on Soil Health and Lower Input Costs. Co-Composting Sewage Sludge with Organic Municipal Waste and Agricultural Residues Further Increases Nutrient Recovery Efficiency. Studies Indicate That Integrated Approaches Improve Both Waste Management and Soil Fertility Outcomes (Guven et al., 2023).

3.4 Decentralized Wastewater Treatment and Recovery Systems

For large numbers of districts in the Narmada Basin, a decentralized approach is best suited. These systems are widely adopted due to their effectiveness and scalability in wastewater treatment. In rural and tribal areas, nature-based solutions systems are better as they require low capital costs and are easy to operate. For small communities, DEWATS are successful systems to enhance water quality by the removal of pollutants. This system also supports wildlife and groundwater recharge (Shi et al., 2022).

4 Investment Plan for Domestic Wastewater Management in the Narmada Basin

Based on district-wise demand vs. capacity gaps, infrastructure status, and identified treatment models, the investment plan for domestic wastewater management in the Narmada Basin has been developed. Based on the basin's heterogeneity model, a base investment strategy is proposed to ensure a proper distribution of funds for the betterment of the environment. Details as shown in *Table 8* and *Figure 5.*

Table 8 : Tier-wise classification of cities (Source: Reserve Bank of India)

Population classification	Population (2001 Census)
Tier-1	100,000 and above
Tier-2	50,000 to 99,999

Tier-3	20,000 to 49,999
Tier-4	10,000 to 19,999
Tier-5	5,000 to 9,999
Tier-6	less than 5,000

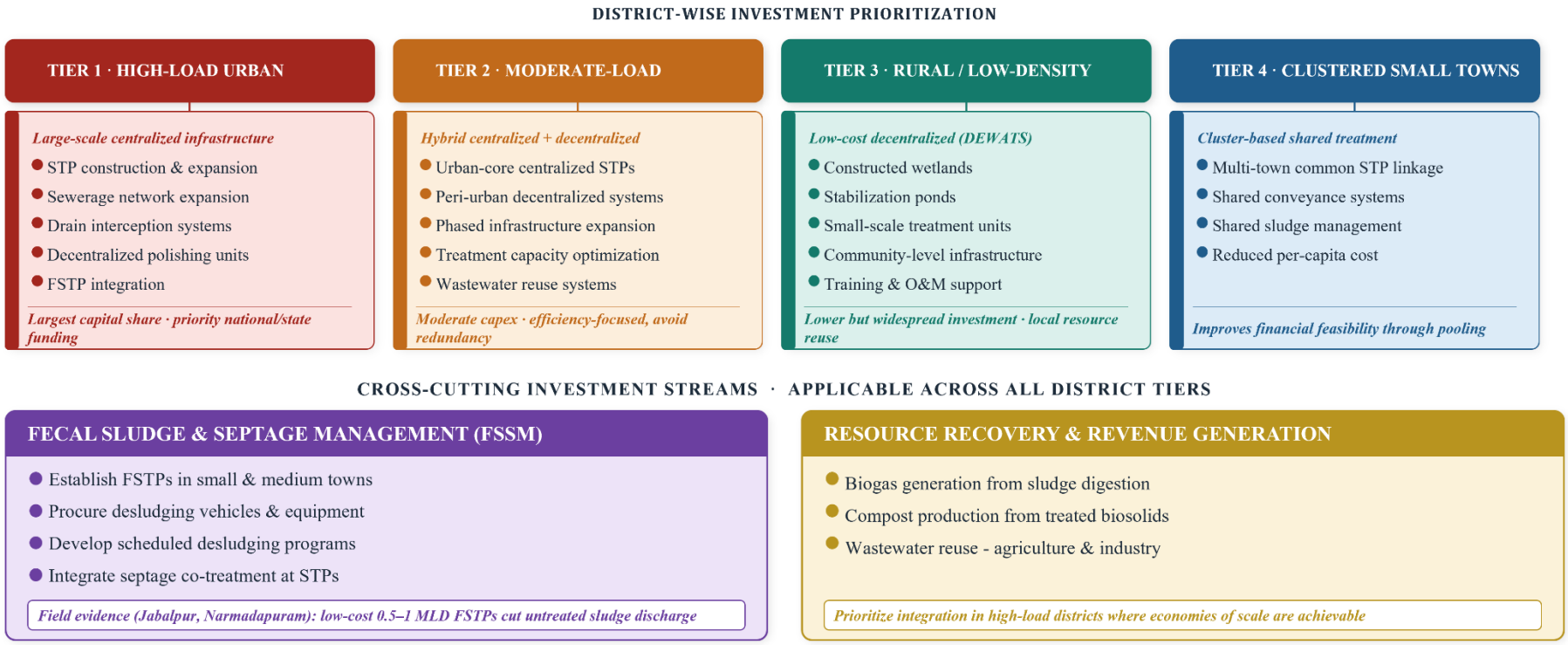


Figure 5: Tiered Investment Allocation Framework for Domestic Wastewater Management in the Narmada Basin - District Prioritization and Cross-Cutting Funding Streams

4.1 Phase-wise Investment Strategy

A phase-wise investment approach is recommended to optimize financial deployment:

- **Phase I (Short-term: 0-5 years):** Focus on high-priority districts with severe deficits, implementation of drain interception systems, and establishment of FSTPs
- **Phase II (Medium-term: 5-10 years):** Expansion of hybrid systems in moderate districts and development of cluster-based infrastructure
- **Phase III (Long-term: 10-20 years):** Full coverage through decentralized systems in rural areas and scaling of resource recovery systems

5 Recommendations

5.1 Budgetary roadmap for domestic wastewater

- Investment distribution in the Narmada Basin should be based on the requirements of the districts, such as the amount of untreated sewage generation, gaps in infrastructure at present, and details about the wastewater treatment technologies to be used.

- There must be proper planning of the funds, which depends upon the population of the district, as districts with more population contribute more to the pollution load and require more funding. On the other hand, medium-density population districts need funding for hybrid wastewater treatment systems.

- The decentralized wastewater treatment System should prioritize rural, low-density districts. DEWATS requires less investment and can cover large geographical areas to effectively remove pollutants.

- Funds should include proper training and technical assistance for communities, operating costs, and maintenance of ongoing treatment units.

- For effective, cost-efficient wastewater management, there should be implementation of a holistic approach. Funds utilization should be done effectively so the maximum pollution can be removed from the environment.

5.2 Government Schemes and Implementation Framework for Narmada Basin

For the successful implementation of domestic wastewater management in the Narmada basin, it is important to strengthen the existing schemes of the state and central governments. Table 9 outlines key schemes, their relevance, and how they can be strategically applied across different district categories.

Table 9 : Details of government schemes applicable to domestic wastewater management in the Narmada Basin.

Scheme	Implementing Ministry	Scope of Intervention	Implementation Strategy
AMRUT 2.0	Ministry of Housing and Urban Affairs	Urban water supply and sewerage	Funding for STPs, sewer networks, drain interception systems
Swachh Bharat Mission – Urban (SBM-U 2.0)	MoHUA	Sanitation, FSSM, waste management	Establishment of FSTPs, desludging systems, septage management
Jal Jeevan Mission (JJM)	Ministry of Jal Shakti	Rural water supply and greywater management	Development of decentralized greywater treatment and reuse systems
Namami Gange Program (Adaptable Model)	Ministry of Jal Shakti	River conservation and pollution control	Interception of drains, riverfront STPs, and monitoring systems
Atal Mission for Rejuvenation (State Schemes)	State Governments	Urban infrastructure development	Hybrid systems, cluster-based STPs
SATAT (Sustainable Alternative Towards Affordable Transportation)	Ministry of Petroleum & Natural Gas	Biogas and CBG production	Biogas generation from sludge, revenue generation
MGNREGA (Rural Employment Scheme)	Ministry of Rural Development	Rural infrastructure development	Construction of wetlands, ponds, decentralized systems
Smart Cities Mission	MoHUA	Urban innovation and infrastructure	Smart monitoring of STPs, reuse systems, IoT-based control

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